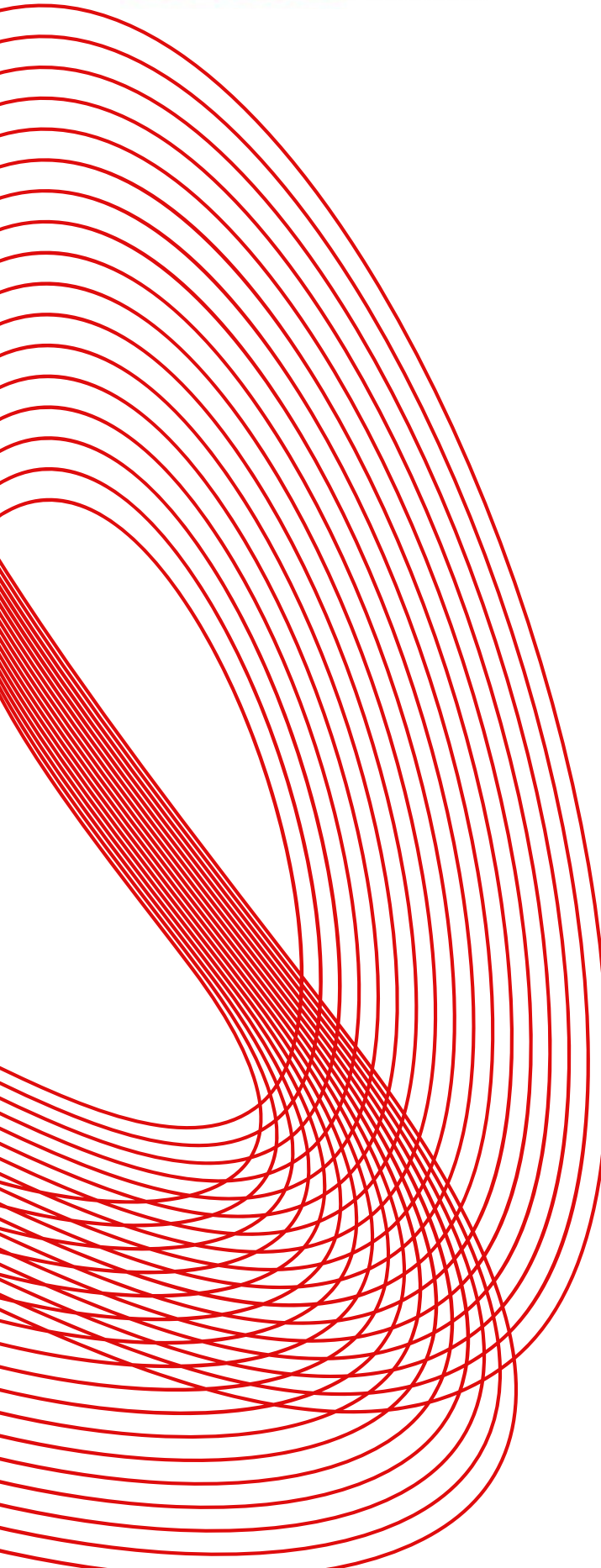


A large, abstract graphic on the left side of the page, composed of numerous concentric, overlapping red lines that form a swirling, tunnel-like shape.

S-IoT Project Report : Drone swarm

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Specifications

Objective

The purpose of this project is to fly a swarm of UAVs (2 to 3 UAVs) and try to lift off a payload using a net.

Functions

- UWB precise location system
- PID Controller for drone stabilization
- UWB Communication for high frequency refreshing data
- Make a simple back and forth keeping the drone formation (following a leader ?)

Deliverable

- Report with all progress, advancements and explanation of technical knowledge
- Functional UWB code
- Methodology and results of calibration of UWB sensors
- Bibliographical Studies
- Video of Drone demo
- Functional PID Corrector
- Functional ESP32 code implementation using Multi-threading

Bibliographical Studies

0.1 UWB

0.1.1 Definitions

The Ultra-Wideband (UWB) is a technology using radio frequency to exchange data created in 1880. The communication is not the only usage, it can also be used to locate, it's only possible by using different modules, those algorithms and technique will be treated in the next part (0.1.2). In majority of cases, UWB is used for interior purpose, otherwise, it can be used for exterior purpose, but it's less shown in scientific paper.

0.1.2 Location algorithm

In order to use the UWB as a location sensor. We have to use specific algorithm, in the followings of the subsection, we will explain and shown 2 diferents location algorithms. Firstly, the two way ranging (TWR) and secondly the double sided two way ranging (DS-TWR). And finally, we will explain TDOA, the most advanced algorithms.

Two way ranging (TWR)

The Two Way Ranging (TWR) algorithm is the easiest and the fastest to implement in a UWB system. In order to explain this algorithm, we will use 2 modules called UWB_1 and UWB_2 .

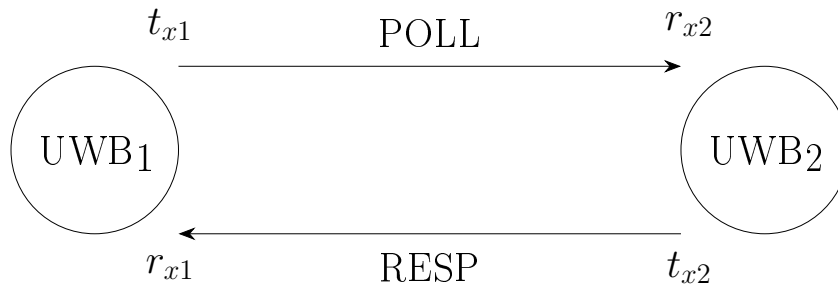


Figure 1: TWR explanation diagram

The algorithm works in 4 steps :

1. UWB_1 sends a **POLL** to UWB_2 . The time of emission t_{x1} is memorized by UWB_1 .
2. UWB_2 receives the **POLL** and memorizes the time of arrival r_{x2} .
3. UWB_2 processes the **POLL**, and precisely plans the time of emission t_{x2} for the **RESP**. It contains (t_{x2}, r_{x2}) .
4. UWB_1 receives the **RESP** and memorizes the r_{x1} .

With all those parameters, we can deduce those variables :

- Total time : $t_{tot} = r_{x1} - t_{x1}$
- Processing time of UWB_2 : $t_{proc} = r_{x2} - t_{x2}$
- Travel time : $t_{travel} = t_{tot} - t_{proc}$

The signal travels back and forth so it did twice the distance (d), and travel speed of a radio signal is equivalent to light speed (c).

So, we can deduce this equation :

$$\frac{t_{travel} \times c}{2} = \frac{(t_{tot} - t_{proc}) \times c}{2} = \frac{((r_{x1} - t_{x1}) - (r_{x2} - t_{x2})) \times c}{2} \quad (1)$$

We can also determine the induced error of this algorithm.

After calculation (describe in calculation part at page 6), we obtain this equation:

$$\epsilon \approx \frac{t_{proc}(e_a - e_b)}{2} \quad (2)$$

As we can observe, the Time of processing is directly link to the error. The t_{proc} is around 3,000,000 nanoseconds so, it's inducing a high error.

Double Sided Two way ranging (DS-TWR)

The DS-TWR has for purpose to reduce the error introduce by the t_{proc} . This algorithm also use UWB_1 and UWB_2 .

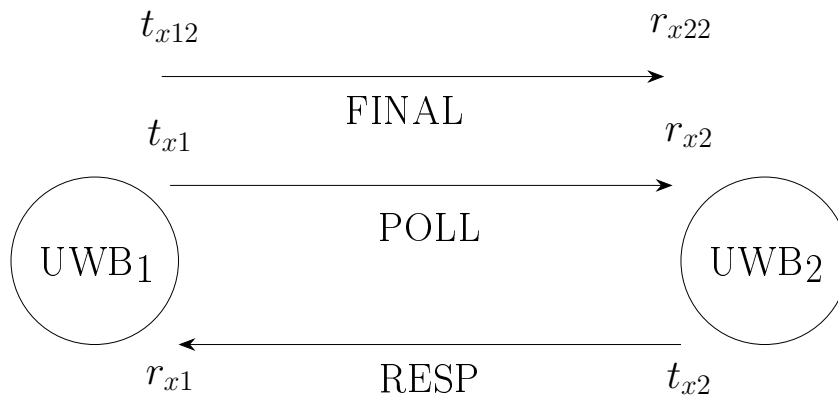


Figure 2: DS-TWR explanation diagram

TDOA

0.2 Measure treatment

In most cases, data measured by sensors are unclear. It could be a sensor offset, it can also come from crosstalk. Or others, exterior interaction. In order, to obtain clean measurements, and to be able to use them for controlling or any other purpose. We have to develop some filter, in this section, we will explain three filters used for cleaning UWB signal, Firstly, we will explain the polynomial filter, and secondly, the Kalman filter, then finally the extended Kalman filter

0.2.1 Polynomial Filter

0.2.2 Kalman Filter

0.2.3 Extended Kalman Filter

Caculations

TWR Error calculation :

Let's set:

- True time of flight : T_f
- True time of process : t_{proc}
- Clock drift errors : e_a and e_b

The measured times are:

$$R_a = (2T_f + t_{proc})(1 + e_a) \text{ and } t_{proc_b} = t_{proc}(1 + e_b) \quad (3)$$

The calculated time of flight is:

$$\hat{T}_f = \frac{R_a - t_{proc_b}}{2} \quad (4)$$

By replacing R_a and t_{proc_b} , we obtain :

$$\hat{T}_f = \frac{2T_f(1 + e_a) + t_{proc}(1 + e_a) - t_{proc}(1 + e_b)}{2} \quad (5)$$

By simplifying, we obtain :

$$\hat{T}_f = T_f(1 + e_a) + \frac{t_{proc}(e_a - e_b)}{2} \quad (6)$$

The error is $\hat{T}_f - T_f$, so:

$$Error = T_f \times e_a + \frac{t_{proc}(e_a - e_b)}{2} \quad (7)$$

The time of flight is about 33 nanoseconds, however D is around 3,000,000.
So we can ignore $T_f \times e_a$ cause $T_f \ll t_{proc}$, and we obtain :

$$Error \approx \frac{t_{proc}(e_a - e_b)}{2} \quad (8)$$

DS-TWR Error calculation :